

Asset pricing: Cash flows and discount rates

MSCF Investments

Fall 2021 Mini 1

- ❑ Supply and demand
- ❑ Cash flows and market discount factors
- ❑ Primer on regression analysis
- ❑ Example: Analysis of COVID crisis and pricing of term structure of dividends

Motivation

- What drives asset prices?
 - Cash flows and discount rates
 - What drives return co-movement?
 - What drives expected returns?
 - What drives return variability?

Asset management comes later in the course.

- Understanding what drives prices & empirical properties of returns helps investors invest better.
- Philosophy:
 - Data analysis alone can lead to overfitting.
 - Given a finite “data budget,” use theory to guide empirical questions to investigate.
 - Want to spend your data budget intelligently.
 - Fundamental drivers vs. leading indicators

Questions

- What are investors paying for when they buy Apple stock?
- Why does Apple's stock price change over time?

Market Summary > Apple Inc

148.60 USD

NASDAQ: AAPL

+121.67 (451.80%) ↑ past 5 years

Closed: Aug 27, 7:59 PM EDT · Disclaimer

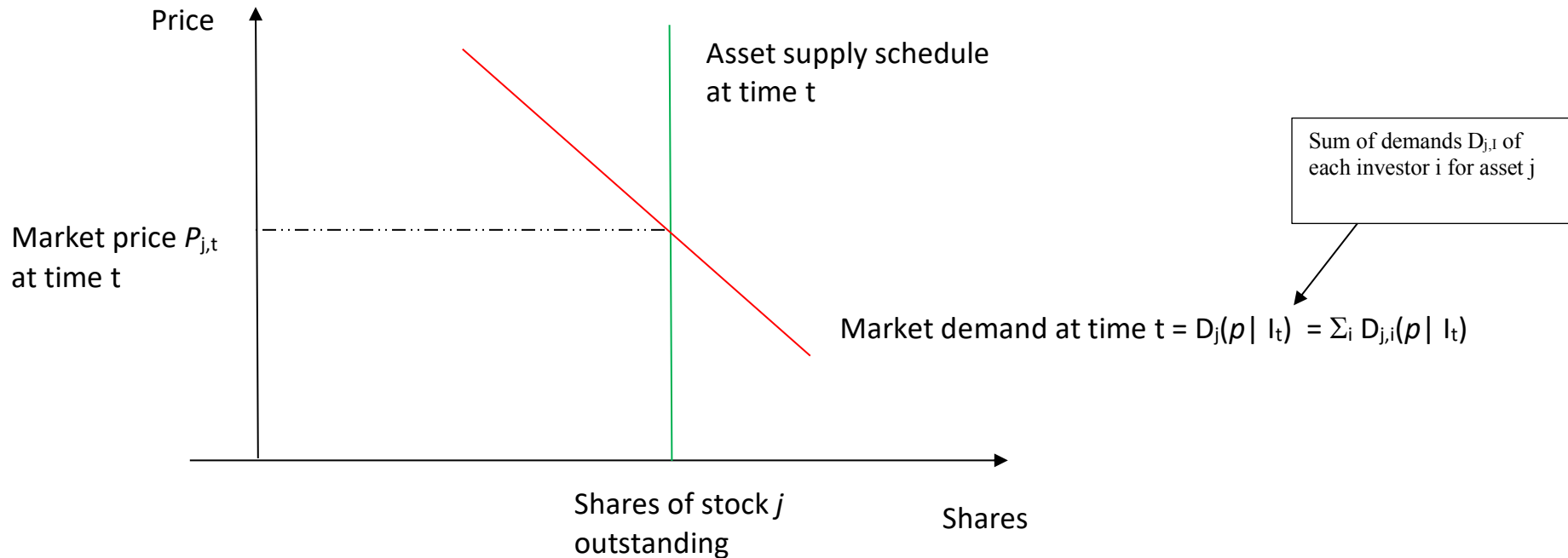
After hours 148.48 -0.12 (0.081%)

1D | 5D | 1M | 6M | YTD | 1Y | 5Y | Max



Open	147.48	Mkt cap	2.46T	Prev close	147.54
High	148.75	P/E ratio	29.10	52-wk high	151.68
Low	146.83	Div yield	0.59%	52-wk low	103.10

Supply and demand for securities



What are investors buying when they buy stock j ?

- Claim against a **stream of future cash flows** (e.g., dividends & retained earnings for a stock)
- Also buying **other characteristics**
 - Industry, index membership, size, financial structure, information structure, location
 - Control rights (i.e., proxy voting)
 - Deliverability for short sells and for options, futures, and other physically settled derivatives
 - Liquidity (i.e., transaction costs)

Prices, expectations, and discounting

Returns: $r_{j,t} = \frac{c_{j,t} + P_{j,t} - P_{j,t-1}}{P_{j,t-1}}$

where $c_{j,t}$ is cash flow paid by j at time t (e.g., dividends, coupons)

Prices as discounted expectations:

$$E_{t-1}[r_{j,t}] = E_{t-1}\left[\frac{c_{j,t} + P_{j,t} - P_{j,t-1}}{P_{j,t-1}}\right] \rightarrow P_{j,t-1} = \frac{E_{t-1}[c_{j,t} + P_{j,t}]}{1 + E_{t-1}[r_{j,t}]} = E_{t-1}[c_{j,t} + P_{j,t}]MDF_{j,t-1}$$

- Expectations E_{t-1} are conditional on **information** I_{t-1} known at time t-1. Change over time. **Why?**
- $E_{t-1}[c_{j,t} + P_{j,t}]$ = valuation **anchor**. Market adjusts price up/down relative to anchor via $MDF_{j,t-1}$.
- **Market discount factor** $MDF_{j,t-1} = \frac{1}{1 + E_{t-1}[r_{j,t}]}$ = effect on price of stock j of its risks & other characteristics and associated market pricing premia
 - Different stocks j may have different **market discount factors** $MDF_{j,t-1}$. **Why?**
 - Discount factors MDFs may change over time. **Why?**
- Timing: When are different variables known?

Prices, expectations, and discounting (2)

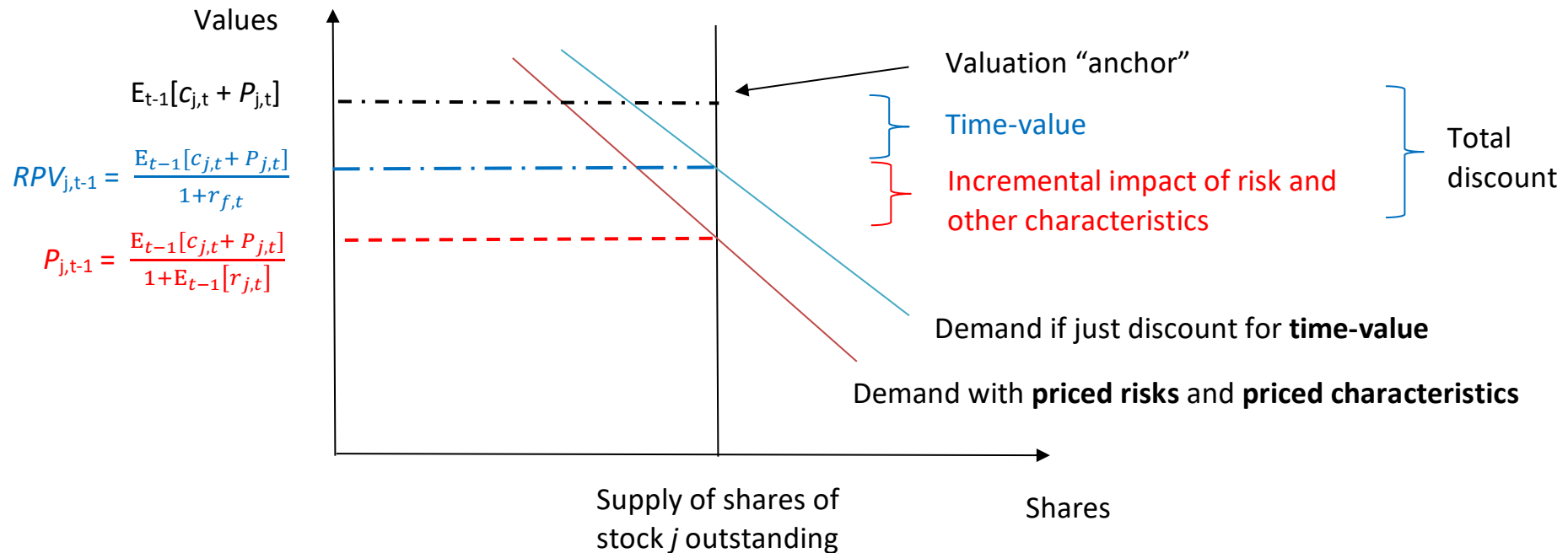
What does $MDF_{j,t-1}$ for stock j depend on at time $t-1$?

$$MDF_{j,t-1} = F(\text{quantity of risk, risk premia, other characteristics, characteristic premia})$$

- **Risk:** Characteristics of prob. distribution from which $c_{j,t} + P_{j,t}$ will be drawn – e.g., variance, skewedness, covariance with other financial and macroeconomic variables (e.g., CAPM beta).
- **Market risk premia:** Market compensation for risk depends on **market-clearing** given aggregate **investor preferences** & total **quantity of risk** that must be borne given outstanding asset supply.
- **Other characteristics** investors value (e.g., low-cost information, good ESG,¹ liquidity, index) or dislike (bad ESG, “sin” stocks, non-index)
- **Market characteristic pricing premia**
- **Special market conditions** – e.g., take-over battles, squeezes.

¹ Environmental, Social, Governance

Prices, expectations, and discounting (3)



Decomposition:

$$E_{t-1}[r_{j,t}] = [\text{compensation for time-value-of-money}] + [\text{effect of priced characteristics}]$$

$$= r_{f,t} + (E_{t-1}[r_{j,t}] - r_{f,t})$$

- $E_{t-1}[r_{j,t}]$ depends on information I_{t-1} at time $t-1$ about stock j 's risk and other characteristics and about the market risk and characteristic premia.

Why are returns random?

Pricing at time $t-1$

$$\begin{aligned}
 P_{j,t-1} &= E_{t-1}[c_{j,t} + P_{j,t}] * MDF_{j,t-1} \\
 &= E_{t-1}[c_{j,t} + E_t[c_{j,t+1} + P_{j,t+1}] * MDF_{j,t}] * MDF_{j,t-1} \\
 &= [E_{t-1}[c_{j,t}] + E_{t-1}[c_{j,t+1} + P_{j,t+1}] E_{t-1}[MDF_{j,t}] + \text{cov}_{t-1}(E_t[c_{j,t+1} + P_{j,t+1}], MDF_{j,t})] * MDF_{j,t-1}
 \end{aligned}$$

Returns at time t

$$\begin{aligned}
 r_{j,t} &= \frac{c_{j,t} + P_{j,t}}{P_{j,t-1}} - 1 \\
 &= \frac{c_{j,t} + E_t(c_{j,t+1} + P_{j,t+1}) * MDF_{j,t}}{E_{t-1}[c_{j,t} + E_t(c_{j,t+1} + P_{j,t+1}) * MDF_{j,t}] * MDF_{j,t-1}} - 1 \\
 &= [1 + E_{t-1}(r_{j,t})] \frac{c_{j,t} + E_t(c_{j,t+1} + P_{j,t+1}) * MDF_{j,t}}{E_{t-1}[c_{j,t} + E_t(c_{j,t+1} + P_{j,t+1}) * MDF_{j,t}]} - 1
 \end{aligned}$$

1. $E(xy) = E(x)E(y) + \text{cov}(x,y)$
2. Iterated expectations:
 $E_{t-1}[E_t[x]] = E_{t-1}[x]$

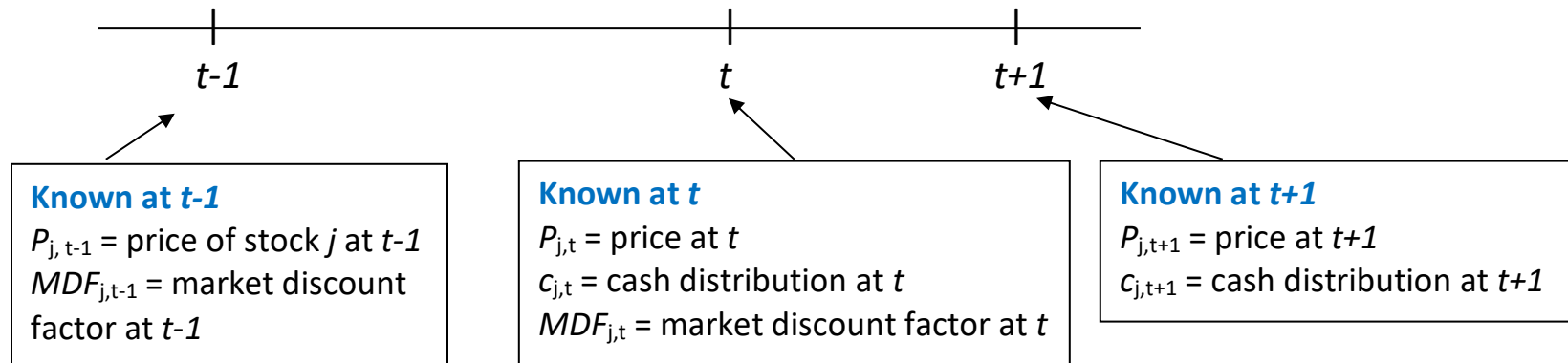
Co-variation of date t expectations of future CFs and prices (i.e., future investment opportunity set) with date t discount factor.

Intuition: Does asset j have high expected cash flows in “states” in which the market values them highly at date t ?

Answer: Because of **news** about **cash flows** and **discount factors** arriving between $t-1$ and t

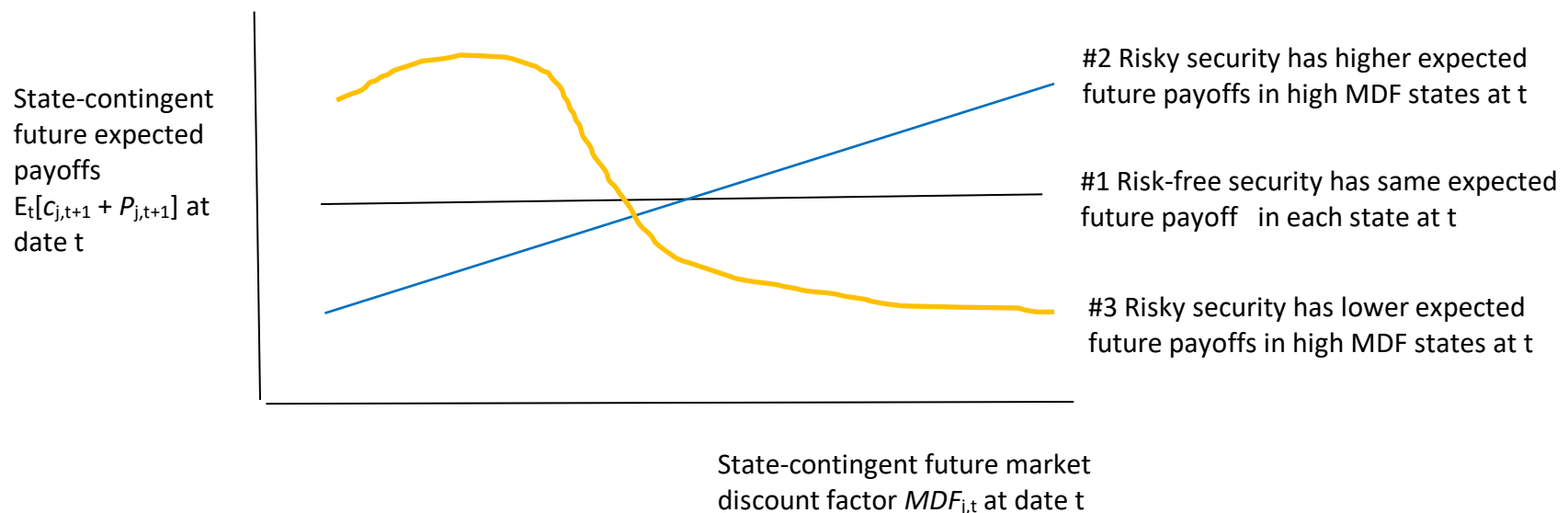
$$c_{j,t} + E_t[c_{j,t+1} + P_{j,t+1}] * MDF_{j,t} \neq E_{t-1}[c_{j,t} + E_t[c_{j,t+1} + P_{j,t+1}] * MDF_{j,t}]$$

Reminder about timing



Covariance pricing intuition

- $\text{cov}_{t-1}(E_t[c_{j,t+1} + P_{j,t+1}], MDF_{j,t})$ imbedded in date t-1 price
- Random changing $MDF_{j,t}$ in different future states at future date t. In some states at date t, market has a high preference for expected future payoffs at t+1 (i.e., a high $MDF_{j,t}$). In others states at t, market has lower (but still positive) preference for expected future payoffs at t+1 (i.e., a low $MDF_{j,t}$).
- Consider 3 different securities with **same future payoff expectations** $E_{t-1}(c_{j,t+1} + P_{j,t+1})$ but **different covariances** of their future expected payoffs with future market discount factor. $MDF_{j,t}$
- Which will have a higher value today?



- **Future question:** What types of future states do investors like more/less at t-1?

When are realized returns high?

Question: When is $r_{j,t} > E_{t-1}(r_{j,t})$?

Formula:

$$r_{j,t} = \left[1 + E_{t-1}(r_{j,t}) \right] \frac{c_{j,t} + E_t(c_{j,t+1} + P_{j,t+1}) * MDF_{j,t}}{E_{t-1}[c_{j,t} + E_t(c_{j,t+1} + P_{j,t+1}) * MDF_{j,t}]} - 1$$

Answer: When CF and MDF news between t-1 and t makes

$$c_{j,t} + E_t[c_{j,t+1} + P_{j,t+1}] * MDF_{j,t} > E_{t-1}[c_{j,t} + E_t[c_{j,t+1} + P_{j,t+1}] * MDF_{j,t}]$$

- Good news about **current dividend**: $c_{j,t} > E_{t-1}[c_{j,t}]$
- Good news about **future cash flows and prices**: $E_t[c_{j,t+1} + P_{j,t+1}] > E_{t-1}[c_{j,t+1} + P_{j,t+1}]$
- Unexpected increase in **next period's market discount factor**: $MDF_{j,t} > E_{t-1}[MDF_{j,t}]$

Question

How much price (or return) variability is due to changing CF expectations vs. changing discount rate?

Old thinking:

- Price changes primarily driven by changing $E[CF]$.
- Discount rates don't change much.

New thinking:

- Changing market discount rates have a significant impact on prices.²

² For example, see John Cochrane (2011), "Presidential Address: Discount Rates," *Journal of Finance*, 1047-1108

Price vs. Cash flow variability

- Empirically, **substantial** time-variation in **both** stochastic MDFs and CFs

P = **Actual** US stock market index value (i.e., given actual MDFs and dividend dynamics)

P_d = **Estimated** market index value holding MDFs **constant** over time

7.2. Present-Value Relations and US Stock Price Behavior

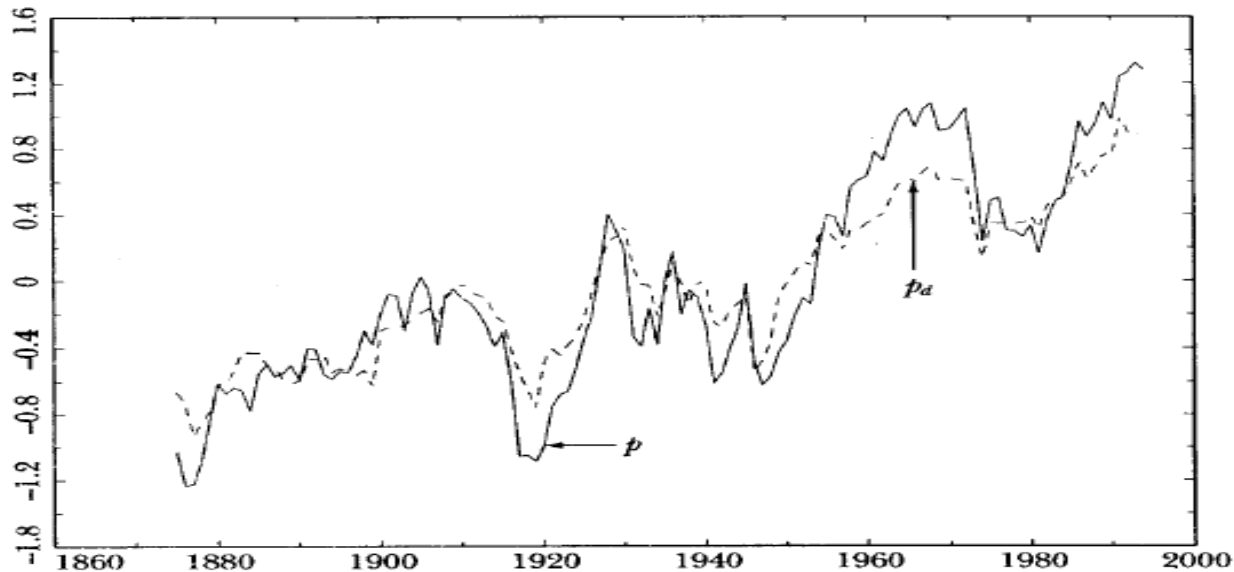


Figure 7.2. Log Real Stock Price and Estimated Dividend Component, Annual US Data, 1876 to 1994

From: *The Econometrics of Financial Markets*, by Campbell, Lo, and MacKinlay

Question: Momentum?

Are high **past** realized returns $r_{j,t}$ between $t-1$ and t a **signal** of high **future** expected returns $E_t[r_{j,t+1}]$?

Answer: Not necessarily! It's complicated.

Possibility #1: Suppose positive return $r_{j,t}$ just due to good CF news but future MDFs – and thus, future $E_t[r_{j,t+1}]$ – are unchanged. In this case, good past CF news doesn't necessarily imply anything about future market $E[\text{ret}]$.

Possibility #2: If **past positive returns** $r_{j,t}$ are due to **increased future MDFs**, then actually signals **lower** future expected returns given $\text{MDF}_{j,t} = \frac{1}{1+E_t[r_{j,t+1}]}$

□ In other words, the causal chain for the high past realized returns might be:

Reduced future expected return $E_t[r_{j,t+1}] < E_{t-1}[r_{j,t+1}]$ → Increased market discount factors $E_t[\text{MDF}_{j,t}] > E_{t-1}[\text{MDF}_{j,t}]$ → High past realized return $r_{j,t}$ between $t-1$ and t

□ Opposite of what at least some people expect when they see high returns.

□ Possible connection with late 1990's bull stock market

Intuitions

There is a fundamental **difference** between characteristics of a security associated with **high prices** vs. characteristics associated with **high expected returns**.

$$P_{j,t} = \frac{E_t[c_{j,t+1} + P_{j,t+1}]}{1 + E_t[r_{t+1}]}$$

Characteristics	Prices	E[ret]
High expected future cash flows?	High	Ambiguous
High risk	Low	High
High liquidity	High	Low
Other desirable characteristics	High	Low

Primer: Regression analysis

Goal: Want to learn about “model of the world” that generates variable realizations

- Data-generating process

- Conjecture: Past empirical **data** produced by an underlying **linear** process

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \dots + \beta_K x_{K,t} + e_t$$

- Key assumption: **Future** realizations will be generated by **same** process as **past** data

- Questions:

- What are best **estimates** $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K$ of the **true** coefficients $\beta_0, \beta_1, \beta_2, \dots, \beta_K$?
- How accurately have the true coefficients been estimated?
- Are estimates biased in some way?
- Are estimates “significantly” different from **hypothesized** values (e.g., 0, 1, or something else)?

Ordinary Least-Squares (OLS) Regression

- **Data:** Sample of N empirical observations $(y_1, x_{1,1}, \dots, x_{K,1}), \dots, (y_N, x_{1,N}, \dots, x_{K,N})$
- **OLS regression:** Compute estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K$ of true $\beta_0, \beta_1, \beta_2, \dots, \beta_K$ to minimized SSEs

$$\text{Min}_{b_0, \dots, b_K} \sum_{t=1}^N [y_t - (b_0 + b_1 x_{1,t} + b_2 x_{2,t} + \dots + b_K x_{K,t})]^2$$

- In words, set $b_0, b_1, b_2, \dots, b_K$ to values $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K$ that make observed data *most likely*
- If residuals e_t have right properties, then OLS is **BLUE** (best linear unbiased estimate) where “unbiased” means $E[\hat{\beta}_k] = \beta_k$ and “best linear” means lowest estimation error $SE[\hat{\beta}_k]$ relative to true coefficient values β_k
- OLS also provides estimates of **standard errors** $SE[\hat{\beta}_k]$ for estimated coefficients

Interpretation

- **Hypothesis testing:** A **t-statistic** $\frac{\text{estimated } \hat{\beta}_k - \text{hypothesized } \beta_k^h}{SE[\hat{\beta}_k]}$ measures how far away an estimate $\hat{\beta}_k$ is from a hypothesized value β_k^h of the true coefficient β_k .
- **Confidence interval:** The likely range in which the true β_k lies given the estimated $\hat{\beta}_k$
- **Regression residual SD(e_t)** measures variability in data that linear model could not explain.
- **Regression R^2** measures $\text{var}(e_t)$ relative to $\text{var}(y_t)$

t-test intuition

Notation

β_k^{true} = **true** value of coefficient assuming stipulated linear model structure is correct

β_k^{hypoth} = **hypothesized** value of true coefficient β_k^{true}

$\hat{\beta}_k$ = OLS **estimator** of true coefficient β_k in assumed linear model

$\hat{\beta}_k^{estimated}$ = **estimated** value of estimator $\hat{\beta}_k$ from regression given data

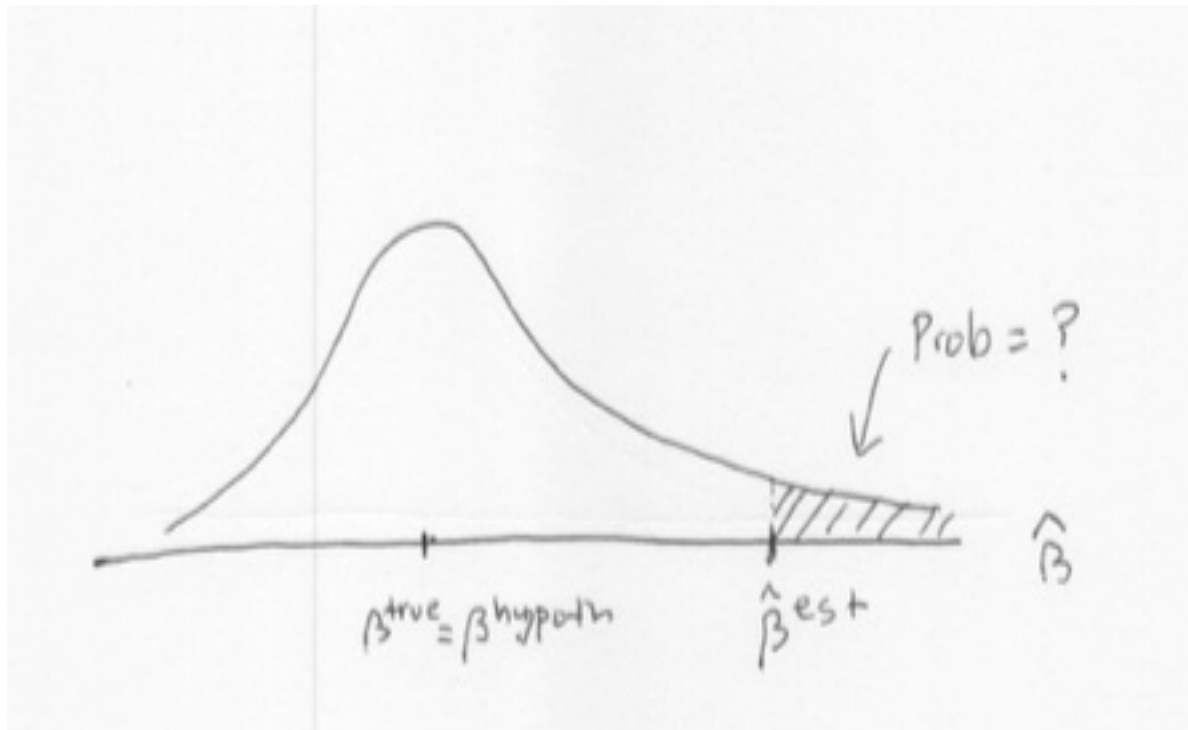
$\hat{t}_k$ = t-statistic **estimator** = $\frac{\hat{\beta}_k - \beta_k^{hypoth}}{SE(\hat{\beta}_k)}$

$\hat{t}_k^{estimated}$ = **estimated** t-statistic value $\frac{\hat{\beta}_k^{estimated} - \beta_k^{hypoth}}{estimated\ SE(\hat{\beta}_k)}$ given data

t-statistic intuition (2)

Classical interpretation of t-test

$$\text{Prob}(\hat{\beta}_k > \hat{\beta}_k^{\text{estimated}} \mid \beta_k^{\text{true}} = \beta_k^{\text{hypoth}}) = \text{Prob}(\hat{t}_k > \hat{t}_k^{\text{estimated}})$$



Example: If $\hat{t}_k^{\text{estimated}} > 1.96$, then $\text{Prob}(\hat{\beta}_k > \hat{\beta}_k^{\text{estimated}} \mid \beta_k^{\text{true}} = \beta_k^{\text{hypoth}}) \approx 2.5\%$

Two-tail p-value $\approx 5\%$

Example: Asset pricing in COVID crisis

Article: Coronavirus: Impact on Stock Prices and Growth Expectations
by Niels J. Gormsen and Ralph S.J. Koijen, working paper: June 9, 2020

Question:

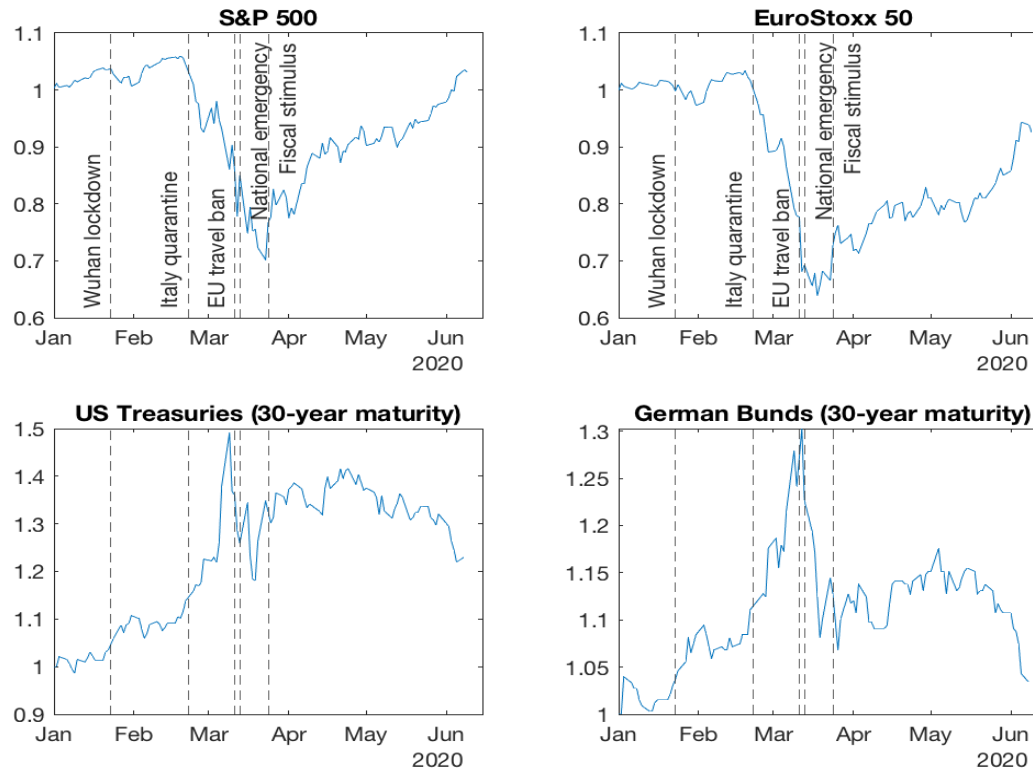
- What are market expectations about how COVID will affect future dividends and future GDP?

Idea:

- Dividend futures are traded for maturities out to 10 years.
- Dividend futures prices depend on current dividends, dividend growth expectations, and current discount rates, but not on post-delivery discount rates. Thus, changes in dividend futures prices are informative about expectations for delivery year.
- Some questions cannot be answered solely by data analytics and historical data. Using models to “read” financial data

Stock and bond market reactions to COVID crisis

Figure 3: The response of the stock and nominal bond markets in the US and EU



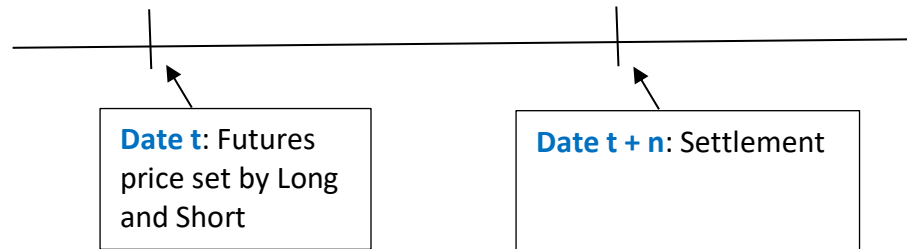
This figure shows the cumulative return on the S&P 500, the Euro Stoxx 50 index, 30-year US Treasuries, and 30-year German bunds. We depict using dashed vertical lines the following five events: The lockdown of Wuhan, China on January 23, the announcement of the quarantine in Italy on February 22, the announcement by the US government that it would ban travel from the EU on March 11, the declaration of national emergency in the US on March 13, and the news that congress is close to passing a stimulus bill on March 24.

Questions

Price changes reflect by impact of COVID on company cash flows and impact on market discount rates.

- How much of the stock market moves are driven by **changes in dividend expectations**?
- How much are driven by **changes in discount rates**? In other words, what is happening to market expected returns for investors willing to hold stock? Was this a good time to buy stock?

Dividend futures



- Traded for S&P 500 on Chicago Mercantile exchange
- Settlement at $t+n$ from Long's viewpoint = $D_{t+n} - F_t^{(n)}$
 - Long received realized dividends D_{t+n} in year $t+n$ from Short
 - Long pays futures price $F_t^{(n)}$ set on day t to Short
- Dividend futures traded on annual dividends over calendar years.
- Paper: Interpolates market dividend futures prices $F_t^{(n)}$ to get estimated constant-maturity prices on an arbitrary day t

Valuation Relations

- Value of stock index

$$S_t = \sum_{n=1, \dots, \infty} P_t^{(n)}$$

where

$$P_t^{(n)} = \frac{E_t[D_{t+n}]}{1 + \mu_t^{(n)}} = D_t \frac{G_t^{(n)}}{1 + \mu_t^{(n)}} = \text{Price on } t \text{ of dividend strip for dividend } D_{t+n} \text{ on future date } t+n$$

$$\mu_t^{(n)} = \text{discount rate on date } t \text{ for pricing a future dividend } D_{t+n}$$

$$G_t^{(n)} = E_t \left[\frac{D_{t+n}}{D_t} \right] = 1 + g_t^{(n)} = E[\text{gross cum. growth rate}] \text{ between } D_t \text{ and future dividend } D_{t+n}$$

- Dividend futures

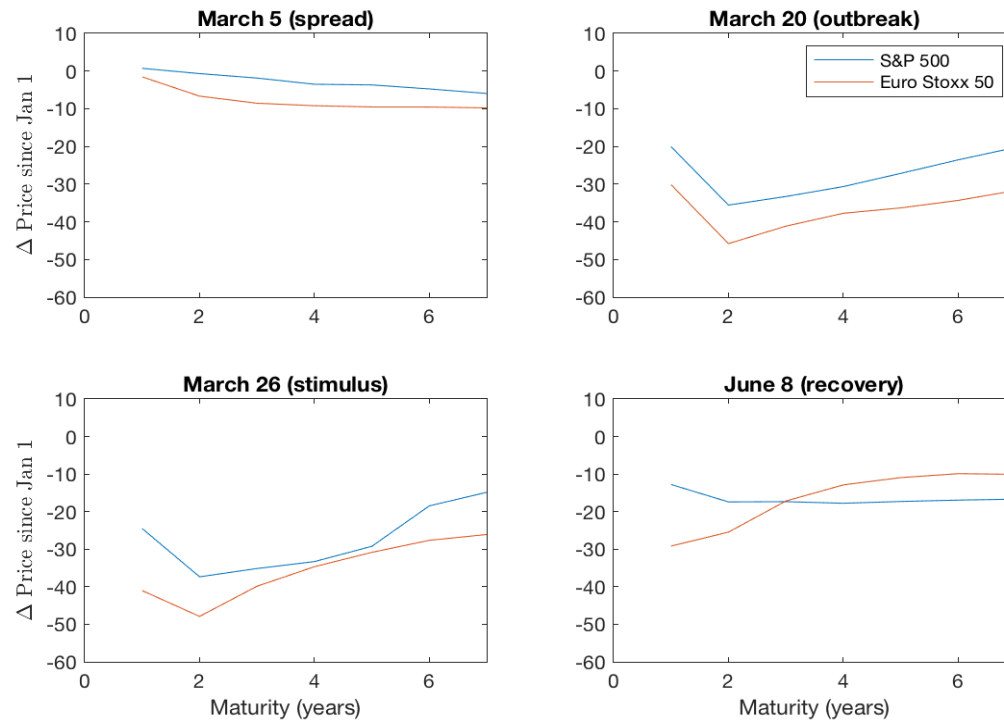
$$F_t^{(n)} = \text{futures price on } t \text{ for dividend } D_{t+n} = P_t^{(n)} (1 + y_t^{(n)}) \rightarrow F_t^{(n)} = D_t G_t^{(n)} \frac{(1 + y_t^{(n)})}{1 + \mu_t^{(n)}} = D_t \frac{1 + g_t^{(n)}}{1 + \theta_t^{(n)}}$$

$$y_t^{(n)} = \text{n-period risk-free interest rate} \rightarrow F_t^{(n)} \approx D_t \left[1 + g_t^{(n)} - (\mu_t^{(n)} - y_t^{(n)}) \right]$$

$$\theta_t^{(n)} = \frac{1 + \mu_t^{(n)}}{1 + y_t^{(n)}} - 1 = \text{n-period expected excess return} \approx \mu_t^{(n)} - y_t^{(n)}$$

Dividend futures term structures and COVID crisis over time

Figure 4: The Development of the Dividend Term Structure over the COVID-19 Outbreak



This figure shows the relative price of dividend of dividend futures with different maturity. We consider the percentage change in prices since January 1. The dividend futures are claims on the dividend paid out on the index in a given year. We consider the S&P 500 index and the Euro Stoxx 50 index. Maturity measured on the horizontal axis is expressed in years. For instance, the top left figure shows that, between January 1 and March 5, dividend futures prices fall by only a few percent for the 2-year claim but by as much as 10% for the 7-year claim.

Dividend prediction model

Definition:

$$e_{i,t}^{(n)} = \frac{1}{n} \ln \left(\frac{D_{i,t}}{F_{i,t}^{(n)}} \right) = \text{Dividend yield for dividend futures} = \frac{1}{n} \ln \left(\frac{1 + \theta_{i,t}^{(n)}}{G_{i,t}^{(n)}} \right) \approx \frac{(\mu_t^{(n)} - y_t^{(n)}) - g_t^{(n)}}{n}$$

Forecasting model:

$$\frac{D_{t+1yr} - D_t}{D_t} = \beta_{i,0}^D + \beta_{i,1}^D e_{i,t}^{(n)} + \epsilon_{t,t+1yr}$$

Data

- Training sample period: 2006 – 2017
- Quarterly observations

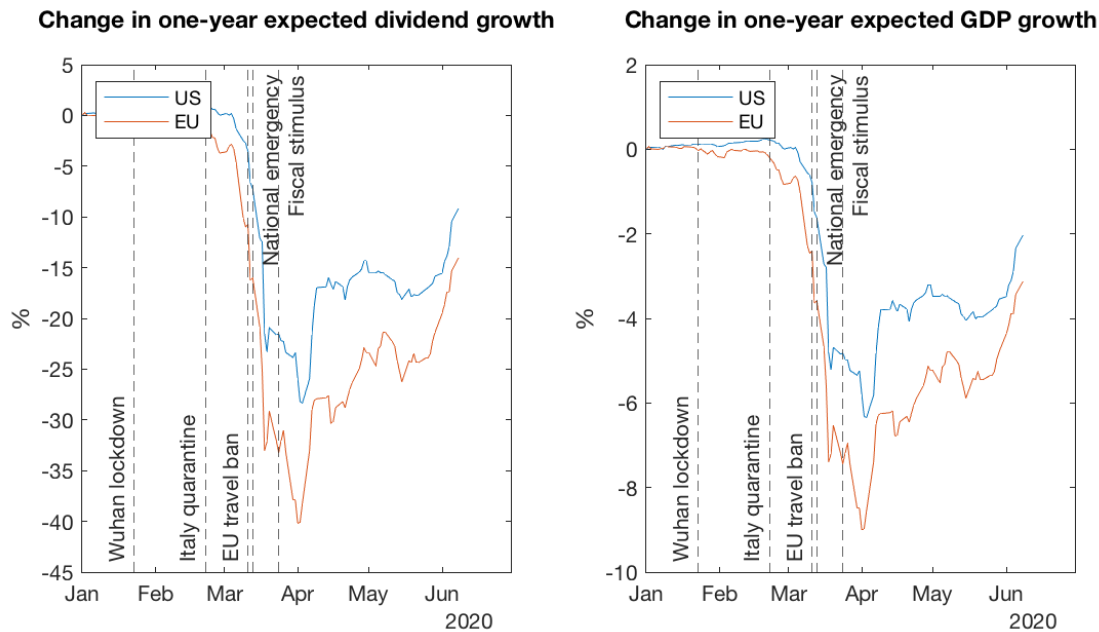
Table 2: Predictive Regressions of Dividend Growth on Dividend Yields

This table shows results from regressions similar to (3). In a pooled sample across S&P 500 and Euro Stoxx 50, we regress realized dividend growth onto the ex-ante two-year yield and a dummy equal to 1 for Euro Stoxx 50 observations. HAC standard errors based on (Lazarus et al., 2019) are presented in parenthesis. Observations are quarterly.

	Intercept	EU dummy	$e_{it}^{(2)}$	R^2	# Obs
$\Delta_1 D_{i,t}$	0.046	-0.018	-0.9518	0.65	98
	(0.02)	(0.03)	(0.19)		

Growth forecasts and COVID crisis

Figure 1: Expected Dividend and GDP Growth from Dividend Futures



This figure shows the change in expected dividend and GDP growth relative to expected value at January 1, 2020. The figure shows expected growth in the US in blue and in the EU in red. Key events are indicated by the vertical dashed lines. The expected dividend growth is revised slowly in response to the outbreak, particularly in the US where it was revised down by less than 5% at March 11. By June 8, expected dividend growth is down by 9% in the US and 14% in the EU. Expected GDP growth over the next year is down by 2.0% in the US and 3.1% in the EU. We emphasize that these numbers are based on historical relations between growth and asset prices and come with uncertainty. Details of the estimation are in Section 5.2.

Learning objectives from COVID paper

- Understand what dividend futures are
- Formula for dividend futures valuation and relation to expected dividend growth, interest rates (time value of money), and expected excess returns (risk premium)
- Can use dividend yield for dividend futures to forecast future dividends and dividend growth

HW: Given dividend forecasts, what are implied expected returns?

Other learning objectives

- ❑ Concept: Supply and demand for securities and market-clearing prices
- ❑ Decomposition: Prices depend on both cash-flow expectations & on market discount factors
- ❑ Intuition: Types of events that cause positive/negative returns vs. high/low prices
- ❑ Intuitions for OLS regression analysis